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Solid State Ionics 138 (2001) 273–276

**SOLID
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Ionic conductivity in the poly(ethylene malonate)/lithium triflate system

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Received 14 August 2000; received in revised form 21 September 2000; accepted 25 September 2000

Abstract

A series of polymer salt complexes based on poly(ethylene malonate) and lithium triflate were prepared with polymer repeat unit to cation ratios of 8, 4, 2, 1. These were characterized and then mixed with lithium triflate. The C=O stretching peaks in the IR spectra of these polymers became broad when salt was introduced, indicating complex formation between the polymer and the lithium cation. Pure poly(ethylene malonate) has a glass transition at 258 K and the glass transition temperature increases with increasing salt concentration. The maximum room temperature ionic conductivity for this polymer/salt complex is 1.6×10^{-6} S/cm for a polymer repeat unit to cation ratio of 8, and the temperature dependence of the conductivity follows the Vogel–Tammann–Fulcher (VTF) equation. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Lithium; Ionic conductivity; Polymer salt complexes

1. Introduction

Solvent-free PEO-based polymer electrolytes are potential electrolyte materials for applications in high-energy batteries and other electrochemical devices [1–4]. Mechanistic studies [5–7] of ion transport in polymer electrolytes demonstrate that the ionic conductivity is enabled by the segmental motion of the polymer hosts and binding energy between cations and anions. These discoveries have been employed to improve the performance of polymer electrolytes. To achieve high segmental motion, flexible polymer backbones with very low glass transition temperatures, such as polyphos-

phazene [8], polysiloxane [9], and substituted alkane [10] are employed. Solvating functional groups were tethered to this backbone to form comb-like polymers. Vibrational spectroscopic studies [11,12] demonstrate the existence of ion pairs in the polymer electrolyte systems and indicate considerable cation–anion interaction. Another approach that improves ionic conductivity is the use of weakly ion-paired salts, such as $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ [13]. Conductivity of poor conductors may be increased by the introduction of cation encapsulating crown ethers, and cryptands [14]. Lewis acids also have been introduced to reduce the basicity of anions and thereby decrease ion pairing [15].

Polymers containing highly polar functional groups have higher dielectric constants and the ability to dissolve more salt than the PEO-based polymers: for example, polycarbonates and poly-

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14.1, δ 41.0, δ 41.3, δ 61.7, δ 166.1 FT-IR (NaCl plate, cm^{-1}): 1751, 1740 ($\nu_{\text{C=O}}$). Anal. Calcd. C, 46.61; H, 4.86; Found C, 46.74; H, 4.79.

Poly(ethylene malonate)/LiCF₃SO₃ polymer salt complexes with different repeating units/cation ratios were prepared by dissolving the polymer and the Li salt in anhydrous THF, followed by the removal of this solvent under rough vacuum, followed by high vacuum (5×10^{-6} Torr) at 100°C for 3 days. Four different materials having repeating unit/cation ratios of 8, 4, 2 and 1 were prepared.

3. Results and discussions

3.1. ^1H NMR spectroscopy

$$\begin{array}{ccccccccccc} \text{CH}_3 & \text{CH}_2 & \text{O} & \left(\text{C} & \text{CH}_2 & \text{C} & \text{O} & \text{CH}_2 & \text{CH}_2 & \text{O} & \text{C} & \text{C} & \text{O} & \text{CH}_2 & \text{CH}_3 \right)_n \\ & & & \parallel & & \parallel & & & & & \parallel & \parallel & & & \\ \text{A} & \text{B} & & \text{E} & & \text{D} & \text{D} & & & & \text{C} & & \text{B} & \text{A} \end{array}$$

Poly(ethylene malonate) was prepared by the polycondensation reaction of ethylene glycol and diethyl malonate in a modification of the method reported by Carothers and Arvin [19]. The preparation of poly(ethylene malonate) terminated by ethoxy groups, was achieved by mixing 5 ml of ethylene glycol (90 mmol) and 14.5 g of diethyl malonate (95 mmol), followed by slowly heating the mixture to 180°C and maintaining that temperature for 4 h, while ethanol was distilled away. The remaining material was heated to 210°C at 0.2 Torr to remove unreacted diethyl malonate. The polymer was then dried on a high vacuum (5×10^{-6} Torr) at 100°C for 3 days and then stored in a closed container in a nitrogen-filled dry box. The product, a pale yellow syrup, was obtained in a yield of 60%. ^1H NMR (CDCl_3): δ 1.26 (t, $J = 7$ Hz), δ 3.39(s), δ 3.43.(s), δ 4.18 (q, $J = 7$ Hz), δ 4.36(s) ^{13}C NMR (CDCl_3): δ

The NMR data indicate a clean polymer product. A peak at 3.39 ppm (C) is assigned to the CH₂ group of the terminated malonate, and a peak at 3.43 ppm (E) is assigned to the CH₂ group of malonate within the repeating units. Integrals of these two peaks indicate that this polymer contains 16 ethylene malonate repeat units indicating a molecular weight of 2260 g/mol.

The FT-IR spectrum of the pure poly(ethylene malonate) contains two strong peaks at 1751 and 1740 cm^{-1} in the C=O stretching region. Additional strong peaks at 1453 and 1416 cm^{-1} are assigned to the CH_2 scissor modes and peaks at 1280 and 1195 cm^{-1} are assigned to the CH_2 twist modes [20]. Lithium triflate salt complexes of the polymer have a broad C=O stretching peak centered at 1745 cm^{-1} . Upon complex formation the two frequencies as-

signed to the CH_2 scissor modes decrease by 5 cm^{-1} while those assigned to the CH_2 twist mode increase by 20 cm^{-1} . These spectral changes are consistent with the observation reported for the coordination between poly(ether esters) and cations [20]. In addition, a shoulder appears at 1615 cm^{-1} when the lithium salt is dissolved in the polymer the intensity of the shoulder increases as the salt content increases, indicating complex formation between the lithium cations and poly(ethylene malonate).

3.3. DSC measurement

Fig. 1a presents the DSC trace of pure poly(ethylene malonate), which is an amorphous material with a glass transition temperature at 258 K. The glass transition temperature is relatively high compared with those of poly(ethylene adipate) 223 K and poly(diethylene carbonate) 246 K. Fig. 1b illustrates

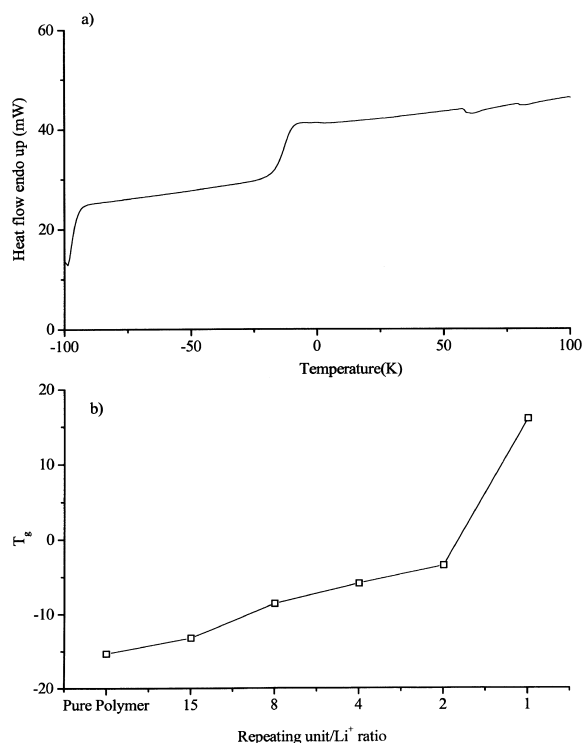


Fig. 1. Thermal analysis of the poly(ethylene malonate)/LiOTf polymer system. a) The DSC trace of the pure polymer. b) Glass transition temperatures change for polymer salt complexes with different repeating unit/cation ratios.

the glass transition temperatures for different polymer/salt complexes. It should be noted that the glass transition temperature increases gradually with increasing salt content, up to a repeat unit/salt ratio equal to 1, and beyond that point the glass transition temperature increases dramatically.

3.4. Ionic conductivity

Ionic conductivities of different poly(ethylene malonate)/lithium triflate complexes are displayed in Fig. 2, where it may be seen that the 8:1 mixture has the maximum ionic conductivity of $4.5 \times 10^{-5}\text{ S/cm}$ at 65°C and $1.6 \times 10^{-6}\text{ S/cm}$ at 25°C . Ionic conductivity decreases as the lithium salt content increases, and this correlates with concomitant increases in the glass transition temperatures of the complexes. The curvature of the conductivity plot reflects the amorphous nature of this salt complex. Accordingly, the conductivity plots fit the Vogel–Tammann–Fulcher (VTF) equation: [21–23]

$$\sigma = AT^{-1/2} \exp(-B/(T - T_0))$$

where T_0 represents an empirical parameter related to the glass transition temperature and is usually 50° lower than T_g and B is a pseudo-activation energy term relating to the expansivity of the polymer complexes. The pre-exponential factor, A , is dependent on the concentration of effective charge carriers.

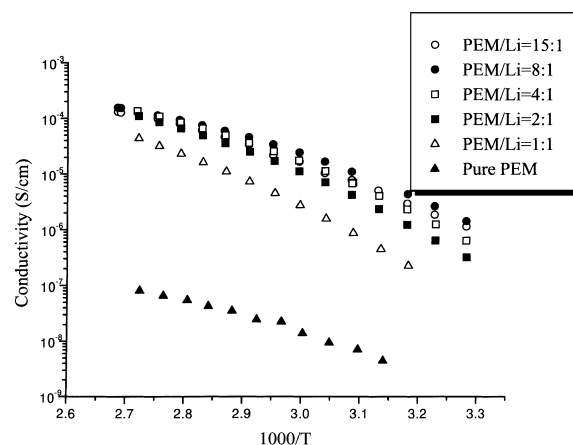


Fig. 2. Ionic conductivity of poly(ethylene malonate)/LiOTf polymer/salt complexes.

Table 1
VTF parameters for poly(ethylene malonate)/LiOTf system

Complex	T_0 (K)	B (K)	A ($\text{SK}^{1/2} \text{ cm}^{-1}$)
PEM/LiOTf=8	215	885.3	0.085
PEM/LiOTf=4	217	1048.2	0.285
PEM/LiOTf=2	220	1124.5	0.437
PEM/LiOTf=1	239	997.7	0.203

As shown in Table 1, the A term increases as the lithium salt content increases, up to the repeating unit/lithium ratio of 2 and then decreases. The B terms follow the similar trend.

The ionic conductivity of poly(ethylene malonate)/lithium triflate is five times higher than that of poly(ethylene adipate)/lithium triflate, but it is the same as that for poly(diethylene glycol carbonate)/sodium triflate. The present poly(ethylene malonate) system has ionic conductivity one order of magnitude lower than that of the aPEO/lithium triflate [24]. Apparently, the polar polymers studied here are good solvents, but the strong interaction between the cations and polymer host stiffens the polymer backbone. Along with the relatively high T_g , this lowers the resulting ionic conductivity.

Acknowledgements

We greatly appreciate the support of this research by ARO award DAAH-95-1-0524.

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